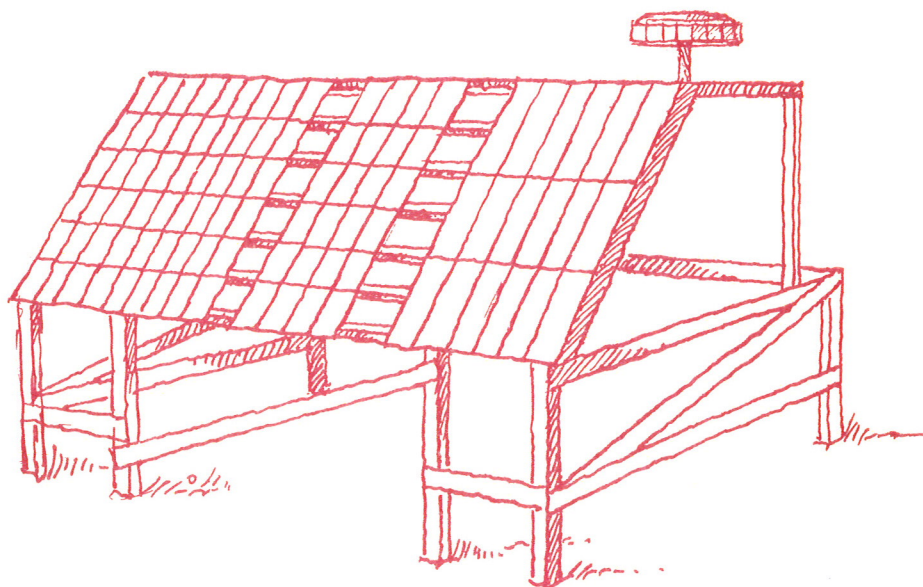


CONVENTION ON LONG-RANGE TRANSBOUNDARY AIR POLLUTION

UN/ECE INTERNATIONAL CO-OPERATIVE PROGRAMME
ON EFFECTS ON MATERIALS, INCLUDING HISTORIC
AND CULTURAL MONUMENTS



Report No 6

Corrosion attack on steel, zinc and aluminium.
Evaluation after 1 and 2 years' of exposure

November 1991

PREPARED BY THE SUB - CENTRE



NATIONAL RESEARCH INSTITUTE
FOR THE PROTECTION OF MATERIALS
PRAGUE/CSFR

UN ECE INTERNATIONAL COOPERATIVE PROGRAMME
ON EFFECTS ON MATERIALS INCLUDING HISTORIC
AND CULTURAL MONUMENTS

Corrosion attack on steel, zinc and aluminium.
Evaluation after 1 and 2 years' of exposure

PREPARED BY THE SUB-CENTER

NATIONAL RESEARCH INSTITUTE FOR THE PROTECTION
OF MATERIALS
190 11 Praha 9 - Běchovice,

P. Holler, D. Knotková and A. Boušková

D E C E M B E R 1 9 9 1

The activity of the sub-center for structural metals is financially supported by the Federal Commission for the Protection of Environment of the Government of Czech and Slovak Federal Republic.

This work is the result of co-operation between the organizations listed below, who participated in the UN ECE exposure programme. Each was responsible for gathering meteorological and pollution data, and for providing sites for the exposure of materials :

National Research Institute for the Protection of Materials,
Czechoslovakia (CSFR)

Technical Research Centre of Finland (VTT), Finland

Bayerisches Landesamt für Denkmalpflege, Germany

Agency for Energy Sources (ENEA), Italy

TNO Division of Technology for Society, Department of
Environmental Chemistry, the Netherlands

Norwegian Institute for Air Research, Norway

Swedish Corrosion Institute, Sweden

Building Research Establishment, Department of Environment,
United Kingdom

Ministerio de Obras Publicas y Urbanismo (MPOU), Spain

Institute of Physical Chemistry, Academy of Science,
the USSR

Institute of Technology, Laboratory of Mineralogy and
Petrology, Portugal

National Research Council of Canada and the Ministries of
the Environment of Canada and Ontario, Canada

United States Environmental Protection Agency, USA

Contents

1. Introduction

2. Experimental

2.1 Test sites

2.2 Measurement of environmental data

2.3 Exposure conditions

2.4 Chemical composition of tested metals

2.5 Exposure and withdrawal of test specimens

2.6 Evaluation of corrosion attack

3. Results and discussion

3.1 Experimental data

3.2 Assessment of quality of data

3.3 Assessment of the mass losses data

3.4 Classification of corrosivity

3.5 Statistical analysis

3.5.1 Correlation analysis

3.5.2 Regression analysis

3.5.3 Grouping of experimental data

4. Conclusions

5. References

1. Introduction

This study is a part of an extensive field exposure programme within the United Nations Commission for Europe (UN ECE), which was started in September 1987 and involves 39 exposure sites in 11 European countries together with the United States and Canada [1].

The aim of the programme is to perform a quantitative evaluation of the effect of sulphur pollutants in combination with NO_x and other pollutants as well as climatic parameters on the atmospheric corrosion of important metallic and non-metallic materials and coatings. This is achieved by measuring gaseous pollutants, precipitation and climate parameters at each test site and by evaluating the corrosion or degradation effects on materials and coatings.

The aim of the work of the subcenter for structural metals - the National Research Institute for the Protection of Materials, CSFR is to perform an evaluation of the effect of pollutants and climatic factors on the atmospheric corrosion of steel, zinc and aluminium.

The present report gives a survey of experimental data from the first two years of exposure, a critical assessment of data obtained and a statistical treatment of the results supporting the main goals of the UN ECE testing programme.

2. Experimental

2.1 Test sites

The programme is undertaken by exposing specimens at test sites with measurements of parameters which are expected to influence significantly the materials and coatings tested. Tab.(1) shows the number and location for 39 test sites participating in the testing programme.

2.2 Measurement of environmental data

At or nearby each site, environmental data are continuously measured. This includes data for gaseous pollutants (SO_2 , NO_2 and O_3), precipitation (mm, pH, SO_4^{2-} as sulphur, NO_3^- as nitrogen, chloride, conductivity, NH_4^+ as nitrogen, Na, Ca, Mg and K) and climatic parameters (temperature, relative humidity, time of wetness and sunshine hours). All climatic measurements are reported to, and compiled by the Norwegian Institute for Air Research, which also checks the quality of the data.

2.3 Exposure conditions

All specimens of structural metals were exposed unsheltered on the racks and sheltered in the box of unified form designed for this testing programme.

The exposure of specimens is in agreement with the ISO 8565.

2.4 Chemical composition of tested metals

Unalloyed carbon steel (C<0.2%, P<0.07%, Cu<0.07%)

Weathering steel ATMOFIX (0.097 %C, 0.60 %Mn, 0.71 %Si, 0.125 %P, 0.013 %S, 0.77 %Cr, 0.51 %Ni, 0.48 %Cu)

Zinc 98.5 %

Aluminium 99.5 %

2.5 Exposure and withdrawal of test specimens

Triplicate test specimens of mentioned materials were exposed from Sept. 1987 to Sept. 1988 (1 year of exposure) and from Sept.1987 to Sept.1989 (2 years of exposure) at two categories of location, i.e. unsheltered and sheltered.

2.6 Evaluation of corrosion attack

The appearance of each specimen was inspected visually or using a stereomicroscope at low magnification (20X). Test specimens from all localities were documented photographically.

The rust layer on weathering steel was evaluated using "rust prints", i.e. the amount of non-adherent rust particles was determined. These rust particles were separated from surfaces of specimens using self-sealing label. The appearance of patina was evaluated using standard scales.

The density and depth of the local corrosion attack on aluminium were evaluated using the methods described in Doc. ISO/TC 156/WG6, N3, i.e. standard charts for the density and size of a local attack and a microscopical method (using a microscope with calibrated fine-focus knob) for the depth of pitting.

Corrosion products were removed from the surface of test specimens in accordance with ISO/DIS 8407.3.

The following cleaning procedures were used :

Steel	method	C3.5
Zinc		C9.5
Aluminium	modified method	C1.1

The interval method of pickling was used for all test specimens. The results then were expressed as mass losses in g/m². The outlying values were eliminated using Dean-Dixon Q-test for $\alpha = .95$.

3. Results and discussion

3.1 Experimental data

The actual set of experimental corrosion data includes the mass losses of the following metals :

Unalloyed carbon steel - 1 year of exposure only (this exposure was intended for the classification of corrosivity according to ISO 9223)

Weathering steel - 1 year, 2 years of exposures

Zinc - 1 year and 2 years of exposures

Aluminium - 2 years of exposure only because the results of 1 year of exposure are not reliable due to inconvenient conditions during pickling procedure.

Results of qualitative or semiquantitative evaluation of the character of corrosion products and local corrosion attack are available for the following materials and exposure periods :

Weathering steel - 2 years of exposure; colour, gloss, structure of the rust layer, "rust prints"

Aluminium - 2 years of exposure; density and size of local corrosion attack

Experimental data mentioned above are summarized in Tables (2) to (5) and shown in Figs.(1) to (5).

The results of the pit depth measurements expressed as an average value are in the range of 10 to 35 μm , the maximum values were found in the range of 45 to 60 μm , especially in case of cracked blisters and open pits. The size and character of pitting on aluminium is evident from the SEM pictures for typical exposure sites (Fig.(6)). Examples of "rust prints" are shown in Fig.(7).

3.2 Assessment of quality of data

The set of experimental data prepared for further processing is influenced by uncertainties having various sources and significance.

The first source of the scatter is due to the use of triplicate specimens. It can be influenced by many factors connected with the character of material of individual specimens, i.e. inhomogenities, contamination, etc. or the character of exposure, e.g. the position of specimen on the test rack, height above the ground, wind stream, biological factors, etc.

The differences caused by these errors do not exceed 10 % in case of unalloyed & weathering steels and 25 % for zinc and aluminium.

The second kind of uncertainty is caused by systematic errors shifting both the experimental data and the calculated average to lower or higher values. These errors have an origin, e.g. in the incorrect treatment of test specimens, but mostly it is very difficult to identify the causes of this kind of errors.

The knowledge of the sources and magnitude of these systematic errors is very important because non-reliable data could affect the correctness of interpretation of results.

The assessment of quality of the UN ECE programme data from the point of view of systematic errors was done by comparing the UN ECE programme data with mass losses of steel and zinc from the ISOCORRAG programme [2], exposed parallelly at the same test localities. The experimental results are listed in Table (6) where the 1 and 2 years' values taken from both testing programmes are put together and compared. The significant differences are evident for 1 year of exposure of zinc, where practically all reported data from test sites of the UN ECE programme are significantly higher than tests performed within the ISOCORRAG programme. The results of two years' withdrawal of zinc do not differ so much for both programmes except Kopisty test site.

3.3 Assessment of the mass losses data

Mass losses of 1 year unsheltered exposure of unalloyed steel are distributed in a very broad range, i.e. from 44.8 g/m² for test site No. 33 Toledo to 557 g/m² for No. 3 Kopisty. The range of results for weathering steel is similar, but the values are about 10 to 20 % lower. The comparison of unalloyed and weathering steel corrosion data is shown at Fig. (8).

Mass losses of sheltered specimens, i.e. specimens where the free access of air is limited, are lower for both unalloyed

and weathering steel than in case of unsheltered specimens for 1 year of exposure. The ratio of mass losses sheltered/unsheltered for unalloyed steel varies in the interval of 0.14 (No. 23 Birkenes) to 0.78 (No. 22 Borregaard) with the mean and the most frequent value of about 0.30. The same ratio for weathering steel is shifted to higher values, i.e. the mean value of this ratio is about 0.50 with the maximum of 0.89 (No. 22 Borregaard). The higher values of this ratio were found for the polluted test sites, especially No. 1, 3, 22, but a general tendency cannot be confirmed.

For 2 years of exposure of weathering steel, the ratio of mass losses for sheltered/unsheltered exposure is significantly higher, the mean value being about 0.6. For polluted test sites No. 3, 1 and 22, the mass loss for sheltered specimens is higher than for unsheltered ones. In this case, the absence of favourable influence of rainfall on the formation of protective patina is evident. It is expected that the same corrosion state of weathering steel will be achieved at other, moderate polluted test sites after a long-term exposure.

This hypothesis could be confirmed by 4 years' withdrawal. The specific corrosion behaviour of both unalloyed and weathering steel under shelter was known later and the experimental results of the UN ECE programme (appearance of corrosion products and "rust prints") show significant differences between the layers of corrosion products on the surface of specimens exposed sheltered and unsheltered.

The ratio of corrosion attack sheltered/unsheltered for zinc is in the interval of 0.3 to 0.85, i.e. corrosion of zinc in sheltered conditions is significantly lower. It seems that wetting of the surface by rain is a very important factor. For example, the test sites with higher amount of precipitation (>1000 mm/y) show this ratio at lower part of the interval (≈ 0.4).

The incorrectness of the procedure used in the pickling of aluminium specimens exposed for 1 year limits the evaluation to the 2 years' exposure only.

The mass losses of aluminium for the unsheltered exposure varies from 0.23 to 3.8 g/m^2 . There is not an evident direct relationship to the pollution by SO_2 and the influence of chlorides seems to be significant, e.g. values for UK test sites No. 30, 27 as well as Sweden (No. 25, 24) and Finland No. 6.

The ratio of the corrosion attack sheltered/unsheltered cannot be explained in a simple way. Most of the test sites, especially those more polluted shows higher corrosion attack under shelter than unsheltered.

The deposition rates of corrosion stimulators on the surface of specimens exposed under shelter is low when compared with unsheltered exposure, but without possibility to be washed out by rain, the disposable amount of these corrosion stimu-

lators increases and consequently the corrosion attack is increasing too.

3.4 Classification of corrosivity

The reason for one year exposure of unalloyed carbon steel was to classify the corrosivity of test sites. The use of the materials (unalloyed low carbon steel, as well as zinc and copper) recommended for the classification purposes in the international standard ISO 9226 [3] allows the estimation of atmospheric corrosivity with categories taken from ISO 9223 [4].

Corrosivity category	Corrosion (g/m ² . year)		
	steel	zinc	copper
C1	<10	<0.7	<0.9
C2	10 - 200	0.7 - 5	0.9 - 5
C3	200 - 400	5 - 15	5 - 12
C4	400 - 650	15 - 30	12 - 25
C5	650 - 1500	30 - 60	25 - 50

The order of the test sites in Table (7) was constructed using a selection based on the mass losses of unalloyed steel after 1 year of exposure.

In our opinion, the classification based on the results of 1 year of zinc exposure is not reliable and the corrosivity referring to zinc is consequently given in brackets.

The result of corrosivity classification is a grouping of the test sites into categories with low (C2), medium (C3), high (C4) and very high (C5) corrosivity.

The evaluation of the corrosivity of the UN ECE programme test sites confirms the fact that except some polluted test sites (No. 22, 1 and 3) and dry clean atmosphere of No. 33 Toledo, the test sites fall into category C3 with medium corrosivity.

It seems that grouping of test sites using the ISO corrosivity classification is not sufficient for a deeper analysis of the influence of wet and dry acid deposition on the corrosion of structural metals.

3.5 Statistical analysis

The aim of the following statistical procedures was to explain the significant role of selected and/or combined factors of environment in the corrosion of tested metals, i.e. unalloyed and weathering steels, zinc and aluminium.

The reduction of the influence of some factors under shelter could contribute to knowledge about different action of wet and dry depositions.

Correlation and regression analysis were performed on the set of experimental (1 and 2 years of exposures) data using mean values (arithmetic average) both for corrosion and environmental data.

3.5.1 Correlation analysis

The results of correlation analysis are summarized in Table (8) as a correlation matrix.

The mass losses in g/m^2 of sheltered and unsheltered specimens after 1 and 2 years of exposure were correlated with the decisive factors of environment, especially with those forming the air pollution complex.

Mass losses of unalloyed and weathering steel after 1 and 2 years of exposure correlate significantly ($R > 0.58$) for both sheltered and unsheltered conditions with SO_2 pollution and in a smaller extent with NO_2 pollution. These results confirm the well known dominant role of SO_2 pollution and a less unambiguous influence of NO_2 as it was published formerly [5].

There are test sites with low SO_2 pollution where relatively high corrosion rates (mass losses) could be explained by high NO_2 mean concentration (test sites No. 28, 18, 36, 39, 19, 30, 21).

Humidity expressed as the time of wetness (TOW defined as time with $\text{R.H.} > 80\%$ at $\text{temp.} > 0^\circ\text{C}$) does not seem to be a significant individual factor, but the combination of humidity and pollution appears more important.

As it is shown below, the combined influence of humidity and pollution has not a simple additive character. It is evident that there is a difference between high humidity episodes accompanied by rain shower, that washing out a part of the corrosion stimulators, and the situation where the surface of metal is wetted by adsorbed water or by acid fog or dew, i.e. conditions that allow the stimulators to diffuse more deeply into the layer of corrosion products.

The high correlation between corrosion attack (mass losses) of unsheltered specimens and the conductivity of rainfall is evident because dissolved corrosion stimulators (sulphates, chlorides, etc.) contribute significantly to the conductivity of rain water.

3.5.2 Regression analysis

The values of correlation coefficients, characterizing the fit between corrosion rate and some selected factors of environment, are so significant that a detailed statistical analysis was carried out to give a quantitative information about the contribution of mentioned environmental factors. These factors, i.e. the pollution by SO_2 and NO_2 together with time of wetness, as significant independent variables,

are considered in the following model :

$$K = A + B_1 [SO_2] + B_2 [NO_2] + B_3 [TOW] \quad (\text{eq.1})$$

where K is the mean mass loss (from triplicate specimens) in g/m^2 after 1 and 2 years of exposure, $[SO_2]$ and $[NO_2]$ the mean concentrations in $\mu g/m^3$, [TOW] the time of wetness in hrs/year and A, B_i are constants.

The combined environmental factors, e.g. ($SO_2 \times TOW$) or ($SO_2 + NO_2$), etc. for a limited reliability of individual factors were not considered.

Other environmental factors, e.g. conductivity and pH of rainfall, sulphates and nitrates in precipitation could not be used for the regression analysis for their dependency on gaseous pollutants as SO_2 and NO_x .

The results of stepwise regression analysis of mass losses of steel and zinc and selected environmental factors are summarized in Tables (9) and (10). Owing to the local character of corrosion attack of aluminium, the regression analysis using 2 years' mass loss data seems to be of no importance [6].

The values of regression coefficients for the contribution of individual environmental factors (SO_2 , NO_2 , TOW) demonstrate the decisive influence of SO_2 pollution. Contributions corresponding to other factors (NO_2 , TOW) seem not to be so significant and also the standard error of calculation of these coefficients in some cases exceeds 100%.

The differences of regression coefficients for sheltered and unsheltered exposures, especially for gaseous pollutants (SO_2 , NO_2) are also connected with reducing the pollution inside the shelter.

Applying the environmental data from open air exposure to the description of the exposure under shelter is responsible for these differences. This fact was confirmed by the parallel measurement of SO_2 deposition rate inside and outside the shelter at the test sites No. 1, 2, 3 (Table (11)).

Relatively low correlation coefficients for the relation between corrosion attack and environmental factors, i.e. SO_2 , NO_2 and TOW, for unalloyed and weathering steel, zinc and aluminium are due to the fact that the surface layer of corrosion products is a "controlling factor" of the transport of active components participating in the corrosion process (O_2 , H_2O , corrosion stimulators) from atmospheric environment to the metal surface.

Corrosion rates calculated using the multilinear model (eq. 1) indicate a good agreement of the model with the reality for the test sites with environmental parameters close to the average value. Significant differences were found in case of marginal test localities from the point of view of

extreme climatic and pollution values.

3.5.3 Grouping of experimental data

The limitation existing for a simple equation which could express the contribution of individual environmental factors or their combination to the magnitude of the corrosion attack, was the reason for another data treatment. The grouping of test sites based on selected criteria (environmental, corrosion rate) seems to be a simple approach to the description of complex influence of the environment on the corrosion of structural metals.

The classification of test sites using the criteria of the International standard ISO 9223 based on mass loss of unalloyed steel after 1 year of exposure seems to be too rough because all test sites practically fall into one group.

The selection of test sites using intervals for SO_2 (<12, 12-40, >40 $\mu\text{g}/\text{m}^3$), for NO_2 (<10, 10-40, >40 $\mu\text{g}/\text{m}^3$) and time of wetness (TOW) (<200, 200-400, >400 hrs/month) has led to many sub-groups not covered, in many cases, by any locality. Taking into account the evident influence of SO_2 and NO_2 especially for low polluted atmospheres, the following criteria were used for the proper selection of test sites.

group	G1	low polluted	$\text{SO}_2 < 20 \mu\text{g}/\text{m}^3$ & $\text{NO}_2 < 20 \mu\text{g}/\text{m}^3$
	G2	moderate polluted	SO_2 20-40 $\mu\text{g}/\text{m}^3$ or $\text{SO}_2 < 20 \mu\text{g}/\text{m}^3$ & $\text{NO}_2 > 20 \mu\text{g}/\text{m}^3$
	G3	medium to heavy polluted	$\text{SO}_2 > 40 \mu\text{g}/\text{m}^3$

Grouping of test sites is shown in Table (12) for unsheltered and in Table (13) for sheltered exposure together with significant environmental factors (SO_2 , NO_2 , TOW) and 2 years' corrosion data.

The explanation of some varieties in the corrosion behaviour of tested metals within a one group as well as overlapping of these groups as the influence of other factors, especially time of wetness, NO_2 and chlorides in rainfall, seems to be reasonable.

The contribution of a high humidity and chlorides to the corrosion attack at test site No. 29 Clatteringshaws Loch and on the other hand low time of wetness for No. 33 Toledo could explain the extreme values inside group G1 with low polluted atmospheres.

A similar situation exists in the group G2 (moderate polluted atmospheres) where the corrosion at U.K. test sites No. 27 Lincoln Cath. and No. 30 Stoke Orchard is influenced by chlorides in rainfall. Low corrosion attack at test sites No. 13 Rome and No. 33 Madrid is connected with low mean value of the time of wetness, i.e. the possibility of the

existence of the electrolyte layer on the surface of metals as a main requirement for the corrosion process is significantly reduced.

In the same way, the low corrosivity of test sites No. 15 Milan and No.38 Steubenville in group G3 is probably caused by dry atmosphere.

Excluding the results from above mentioned specific localities increases the correlation between corrosion and SO_2 pollution and the influence of SO_2 pollution as the controlling factor of corrosion can be described by a simple linear equation.

The deviations from this typical relationship are than expressed in terms of partial effects of other environmental factors.

Figures (9) to (11) show the relationship between mass losses of weathering steel, zinc and aluminium and mean values of SO_2 pollution, the outlying specific localities with the significant influence of other environmental factors are evident from the plots corrosion vs. SO_2 .

4. Conclusions

The conclusions from the statistical treatment of data can be expressed as follows :

- for all metals tested and for both categories of location, the SO_2 concentration appears to be the controlling factor,
- even in the presence of SO_2 pollution there are several factors lowering the corrosion rate; the most important one is low time of wetness as may be seen from the results in Madrid, Toledo, Rome, Steubenville,
- specific combinations of factors such as high wetness, high NO_2 pollution, or the presence of chlorides lead in specific cases to an unexpected increase of the corrosion rate (Clatteringshaws Loch, Borregaard, Lincoln Cath.),
- no definite conclusion can be drawn about the effect of NO_2 pollution; besides taking part in increasing the high corrosivity in the presence of higher levels of SO_2 pollution, it seems that at low SO_2 the effect of NO_2 becomes more important,
- the results do not yet allow to establish any definite idea about the synergistic effect of SO_2 and NO_2 for a long-term corrosion of structural metals,
- it has no technical sense to derive equation for the damage functions based on the results of 2 years of exposure for weathering steel and aluminium because a steady state

of corrosion has not yet been reached; derived relationships would have the validity limited on the initial period of corrosion and could be used only for an assessment of the significance of environmental factors within this time period,

- it seems not possible to specify target or critical pollution levels on the available results of 2 years of exposure; to derive such levels, it will be necessary to acquire a better knowledge of a reliable corrosion rate prediction method,
- taking into account the decisive influence of SO₂, it can be recommended to use the mean values of SO₂ pollution if critical and target levels for structural metals would be derived.

5. References

- [1] Kučera V., Coote A.T., Henriksen J., Knotková D., Leygraf C., Reinhardt U. : Proc. 11th Int.Congress Metal.Corros., Florence, 2, 433 (1990)
- [2] ISO/TC 156/WG 4, N 211 : ISOCORRAG - International co-operative programme - Survey of results, 1991
- [3] ISO 9226 Corrosion of metals and alloys. Corrosivity of atmospheres. Method of determination of corrosion rate of standard specimens for the evaluation of corrosivity
- [4] ISO 9223 Corrosion of metals and alloys. Classification of corrosivity of atmospheres
- [5] Kučera V., Knotková D., Gullman J., Holler P. : Proc. 10th Int.Congress.Metal.Corros., Madrás, 1, 167 (1988)
- [6] Graedel T.E. : J. Electrochem. Soc. 136 (4), 204 (1989)

Tab.1 Test sites of the UN ECE testing programme

No.	Test site	Country	Environment
1	Praha	Czechoslovakia	U
2	Kašperské Hory		R
3	Kopisty		I
4	Espoo	Finland	U
5	Ahtari		R
6	Helsinki		I
7	Waldhof-Langenbrugge	Germany	R
8	Aschaffenburg		U
9	Langenfeld-Reusrath		R
10	Bottrop	Italy	I
11	Essen-Leithe		R
12	Garmisch-Partenkirchen		R
13	Rome	Italy	U
14	Casaccia		R
15	Milan		U
16	Venice	Netherlands	U
17	Vlaardingen		I
18	Eibergen		R
19	Vredepeel	Norway	R
20	Wijnandsrade		R
21	Oslo		U
22	Borregaard	Sweden	I
23	Birkenes		R
24	Stockholm S		U
25	Stockholm C	United Kingdom	U
26	Aspvreten		R
27	Lincoln Cath.		U
28	Wells Cath.	Spain	U
29	Clatteringshaws Loch		R
30	Stoke Orchard		I
31	Madrid	Soviet Union	U
32	Bilbao		U
33	Toledo		R
34	Moscow	Estonia	U
35	Lahemaa		R
36	Lisbon-Jeronimo Mon.		U
37	Dorset	Canada	R
38	Steubenville		I
39	Res. Triangle Park		R

R - rural
U - urban
I - industrial

Tab. 2 Mass losses of structural metals in g/m² after
1 year of exposure

No.	Test site	Unsheltered			Sheltered		
		Unalloyed steel	Weather. steel	Zinc	Unalloyed steel	Weather. steel	Zinc
1	Praha-Letnany	438	250	14.8	195	190	11.8
2	Kasperské Hory	224	150	8.9	86.9	96.0	7.7
3	Kopisty	557	304	16.2	254	227	10.5
4	Espoo	271	197	11.6	83.7	100	8.4
5	Ahtari	132	90.6	8.9	37.4	56.0	8.4
6	Helsinki	273	243	14.4	114	122	14.0
7	Waldhof-Langenbrugge	264	218	20.9	106	118	20.6
8	Aschaffenburg	213	167	21.3	50.3	71.0	18.2
9	Langenfeld-Reusrath	293	235	28.7	82.9	101	24.0
10	Bottrop	373	257		86.2	111	21.0
11	Essen-Leithe	342	243	30.1	118	118	14.8
12	Garmisch-Partenkirch.	133	112	8.3	44.2	58.0	4.5
13	Rome	178	154	7.7	49.2	59.7	5.2
14	Casaccia	235	185	8.8	97.9	98.5	5.1
15	Milan	366	279	11.3	101	92.0	7.8
16	Venice	245	193	13.5	71.4	95.0	5.6
17	Vlaardingen	344	272	14.4	115	133	7.7
18	Eibergen	232	193	9.9	55.0	73.7	9.0
19	Vredepeel	283	220	16.5	100	119	10.2
20	Wijnandsrade	259	199	20.1	56.0	76.4	12.4
21	Oslo	229	188	27.5	46.0	60.9	24.7
22	Borregaard	430	322	43.4	335	272	38.4
23	Birkenes	194	163	27.4	26.7	39.5	14.0
24	Stockholm S	264	238	10.3	92.5	92.0	8.2
25	Stockholm C	263	244	9.2	66.8	73.9	7.1
26	Aspvreten	147	132	8.2	43.4	52.5	5.0
27	Lincoln Cathedral \	315	289	10.4	155	144	7.1
28	Wells Cathedral	253	214	10.0	85.3	99.0	5.8
29	Clatteringshaws Loch	235	190	12.8	68.0	85.5	5.4
30	Stoke Orchard	307	246	15.1		145	8.2
31	Madrid	222	169	7.9	52.3	66.8	5.9
32	Bilbao	322	218	13.1	76.4	77.6	5.5
33	Toledo	44.8	54.7	3.3	13.1	10.1	2.0
34	Moscow	181	169	10.5	48.3	69.3	9.9
35	Iahemaa	185	169	7.1	89.3	102	5.7
36	Lisbon-Jeronimo Mon.	224	189	9.8	69.0	76.8	5.3
37	Dorset	149	123	9.8	46.5	62.9	5.3
38	Steubenville	214	213	13.5	72.8	106	6.1
39	Res.Triangle Park	176	160	10.6	79.2	93.4	6.1

Tab. 3 Mass losses of structural metals in g/m² after 2 years of exposure

No.	Test site	Unsheltered			Sheltered		
		Weather. steel	Zinc	Aluminium	Weather. steel	Zinc	Aluminium
1	Praha-Letnany	361	26.1	0.91	365	13.3	1.71
2	Kasperské Hory	236	14.0	0.48	180	8.8	0.90
3	Kopisty	427	29.6	1.25	476	14.3	1.98
4	Espoo	275	15.5	0.71	158	10.1	0.68
5	Ahtari	159	8.8	0.48	97.0	6.7	0.35
6	Helsinki	356	20.6	1.69	205	15.1	1.57
7	Waldhof-Langenbrugge	315	26.9	0.56	190	21.6	0.60
8	Aschaffenburg	262	40.1	0.56	133	22.2	0.56
9	Langenfeld-Reusrath	334	42.6	0.96	182	31.8	1.07
10	Bottrop	380			201	20.2	0.96
11	Essen-Leithe	342	23.4	1.36	204	9.8	2.76
12	Garmisch-Partenkirch.	162	12.6	0.29	115	5.6	0.28
13	Rome	204	11.8	0.40	106	7.7	0.75
14	Casaccia	244	14.3	0.37	146	7.0	0.85
15	Milan	316	23.1	0.96	206	10.6	1.35
16	Venice	273	18.3	0.54	161	9.3	1.21
17	Vlaardingen	347	25.9	1.11	230	18.3	1.92
18	Eibergen	263	12.6	0.42	127	5.8	0.40
19	Vredepeel	317	18.5	0.63	173	7.4	1.05
20	Wijnandsrade	278	20.0	0.49	134	6.5	0.49
21	Oslo	242	16.2	0.99	121	7.1	0.81
22	Borregaard	430	48.3	3.80	510	25.4	5.30
23	Birkenes	220	20.5	0.48	103	12.1	0.34
24	Stockholm S	299	23.2	1.16	150	19.8	0.98
25	Stockholm C	329	14.4	1.01	123	7.5	0.90
26	Aspvreten	180	9.0	0.46	105	5.7	0.28
27	Lincoln Cathedral	424	17.0	2.73	251	10.1	2.86
28	Wells Cathedral	280	16.0	0.77	162	7.8	1.59
29	Clatteringshaws Loch	267	23.7	1.06	223	7.0	1.32
30	Stoke Orchard	333	23.1	1.75	235	9.5	2.69
31	Madrid	199	9.6	0.40	109	8.0	0.45
32	Bilbao	295	25.2	0.87	118	7.8	1.22
33	Toledo	93.1	4.5	0.23	50.3	2.3	0.21
34	Moscow	250	14.0	0.39	120	10.8	0.59
35	Lahemaa	212	9.5	0.43	156	6.2	1.09
36	Lisbon-Jeronimo Mon.	256	14.2	0.86	141	6.2	1.34
37	Dorset	173	14.2	0.84	118	6.0	0.74
38	Steubenville	280	20.1	1.10			
39	Res.Triangle Park	218	19.0	0.54	176	7.4	0.92

Colour	1-orange-brown	Structure	1-coarse particles
	2-light brown		2-adherent rust
	3-medium brown		3-powdered form
	4-dark brown	Gloss	1-glossy
			2-matte

18

Tab. 5 Local corrosion attack of aluminium after 2 years of exposure (evaluation of density and size of pitting according to ASTM G46 standard scale)

No.	Test site	Density		Size	
		unsh.	sh.	unsh.	sh.
1	Praha	5/4	5/3	1	2
2	Kašperské Hory	4/3	5/4	1	1/2
3	Kopisty	3	3	3/4	3
4	Espoo	5/4	3/5	2/3	3/2
5	Ahtari	5/4	4/3	2/3	2/1
6	Helsinki	5/4	3/4	2	3
7	Waldhof-Langenbrugge	3/5	5/4	1/2	1/2
8	Aschaffenburg	3/4	3/4	1/3	1/2
9	Langenfeld-Reusrath	4/5	5	1/2	1/2
10	Bottrop	4/5	5/4	2/3	1/2
11	Essen-Leithe	5/4	5	1/2	1/2
12	Garmisch-Partenkirchen	4/5	5	2/1	1
13	Rome	5/4	5	1/2	1
14	Casaccia	4/5	5	1/2	1
15	Milan	4	4	1/2	2
16	Venice	4/5	4	1/2	1/2
17	Vlaardingen	4/5	5	1/2	1/2
18	Eibergen	3	5	1/2	1
19	Vredepeel	5/4	5	1/2	1/2
20	Wijnandsrade	5/4	5	2	1/2
21	Oslo	5/4	5/4	3/2	1
22	Borregaard	5	5	2/3	3/2
23	Birkenes	4/5	3/5	1/2	1/2
24	Stockholm S	4/5	5/4	2/1	2
25	Stockholm C	5	4	2/1	1/2
26	Aspvreten	4/5	4	1/2	1/2
27	Lincoln Cath.	4/5	5/4	1/2	1/2
28	Wells Cath.	4/5	5	1/2	1/2
29	Clatteringshaws Loch	5	5	1/2	1
30	Stoke Orchard	5/4	5	2	1/2
31	Madrid	4	4	1/2	1/2
32	Bilbao	5	5	1	1
33	Toledo	4/5	3/5	1/2	1
34	Moscow	5/4	4	1/2	1/2
35	Lahemaa	5	4	1/2	1/2
36	Lisbon-Jeronimo Mon.	5	5/4	1	1
37	Dorset	4/5	5	1/2	1/2
38	Steubenville	5	5	1/2	1
39	Res. Triangle Park	4		1	

Tab. 6 Comparison of corrosion data from parallelly exposed steel, zinc and aluminium specimens within the UN ECE (b) and ISOCORRAG (a) testing programmes

Test site	Corrosion in (g/m ²)							
	1 year				2 years			
	Steel		Zinc		Aluminium		Zinc	
	a	b	a	b	a	b	a	b
Kasp.Hory	204	224	13.6	8.9	1.65	0.48	16.9	14.0
Kopisty	556	557	24.9	16.2	3.1	1.25	59.4	29.6
Helsinki	262	273	9.1	14.4	1.24	1.69	17.0	20.6
Espoo	201	271	6.3	11.6	0.38	0.71	11.4	15.5
Ahtari	109	132	5.0	8.9	0.11	0.48	8.5	8.8
Oslo	198	229	9.1	27.5	1.08	0.99	18.5	16.2
Borregaard	485	430	27.4	43.4	2.7	3.8	49.7	48.3
Birkenes	155	194	16.4	27.4	0.54	0.48	21.3	20.5
Madrid	218	222	4.2	7.9	0.32	0.40	9.8	9.6
Res.Tri.Park	182	176	6.0	10.6				

Tab. 7 Classification of corrosivity of test sites included in the UN ECE testing programme based on corrosion data for unalloyed steel

No. Test site	Corrosion (g/m ² .year)		Pollution (ug/m ³)	
	steel	zinc [C]	SO ₂	NO ₂
<u>Corrosivity C2</u>				
33 Toledo	44.8	3.3 [C2]	3.7	8.8
<u>Corrosivity C3</u>				
5 Ahtari	132	8.9 [C3]	5.8	5.5
12 Ga-Pa	133	8.3 [C3]	9.4	12.1
26 Aspvreten	147	8.2 [C3]	3.0	4.9
37 Dorset	149	9.8 [C3]	3.3	1.6
39 Res.Tri.Park	176	10.6 [C3]	9.6	26.3
13 Rome	178	7.7 [C3]	28.6	69.0
34 Moscow	181	10.5 [C3]	19.2	75.0
35 Lahemaa	185	7.1 [C3]	0.5	1.0
23 Birkenes	194	27.4 [C4]	1.3	4.0
8 Aschaffenburg	213	21.3 [C4]	23.7	33.2
38 Steubenville	214	13.5 [C3]	58.1	41.8
31 Madrid	222	7.9 [C3]	18.2	23.9
36 Lisbon	224	9.8 [C3]	6.9	37.6
2 Kasp.Hory	224	8.9 [C3]	19.7	17.9
21 Oslo	229	27.5 [C4]	14.3	51.7
18 Eibergen	232	9.9 [C3]	10.1	23.1
29 Clatteringshaws	235	12.8 [C3]	4.9	3.5
14 Casaccia	235	8.8 [C3]		
16 Venice	245	13.5 [C3]	18.5	30.0
28 Wells Cath.	253	10.0 [C3]	7.2	21.1
20 Wijnandsrade	259	20.1 [C4]	14.3	28.5
25 Stockholm C	263	9.2 [C3]	19.6	46.6
24 Stockholm S	264	10.3 [C3]	16.3	26.6
7 Waldhof	264	20.9 [C4]	15.0	11.9
4 Espoo	271	11.6 [C3]	18.8	21.6
6 Helsinki	273	14.4 [C3]	20.8	31.2
19 Vredepeel	283	16.5 [C4]	12.6	29.3
9 Langenfeld	293	28.7 [C4]	24.8	42.8
30 Stoke Orchard	307	15.1 [C4]	14.8	81.6
27 Lincoln Cath.	315	10.4 [C3]	17.9	68.6
32 Bilbao	322	13.1 [C3]	35.2	34.7
11 Essen	342	30.1 [C5]	30.5	46.8
17 Vlaardingen	344	14.4 [C3]	34.8	51.3
15 Milan	366	11.3 [C3]	79.9	122
10 Bottrop	373		50.7	47.9
<u>Corrosivity C4</u>				
22 Borregaard	430	43.4 [C5]	35.8	18.6
1 Letnany	438	14.8 [C3]	77.5	42.4
3 Kopisty	557	16.2 [C4]	83.3	42.2

Tab. 8 Correlation matrix for mass losses of structural metals and environmental factors
(Ws - weathering steel, Fe - unalloyed steel,
ush - unsheltered, sh - sheltered)

		1	2	3	4	5	6	7	8	9
1	Ws/2y/ush	x	.77	.68	n	.73	n	.58	.43	.31
2	Ws/2y/sh		x	n	.47	n	.80	.67	.14	.00
3	Zn/2y/ush			x	.86	.56	n	.44	.19	.18
4	Zn/2y/sh				x	n	.36	.35	.17	.14
5	Al/2y/ush					x	.92	.30	.19	.13
6	Al/2y/sh						x	.40	.22	.14
7	SO ₂							x	.58	n
8	NO ₂								x	n
9	TOW									x
10	S prec.									
11	Cl prec.									
12	pH									
13	Conductivity									
14	Fe/1y/ush									
15	Fe/1y/sh									
16	Ws/1y/ush									
17	Ws/1yr/sh									

		10	11	12	13	14	15	16	17
1	Ws/2y/ush	.36	.41	.27	.67	n	n	.97	n
2	Ws/2y/sh	n	n	n	n	n	n	n	.97
3	Zn/2y/ush	.27	.37	.37	.58	n	n	n	n
4	Zn/2y/sh	n	n	n	n	n	n	n	n
5	Al/2y/ush	.10	.29	.38	.45	n	n	n	n
6	Al/2y/sh	n	n	n	n	n	n	n	n
7	SO ₂	.68	n	.21	.60	.76	.56	.61	.57
8	NO ₂	n	n	.04	n	.42	.15	.51	.19
9	TOW	n	n	n	n	.13	.00	.29	.00
10	S prec.	x	n	.18	.54	.57	n	.41	n
11	Cl prec.		x	n	.49	.40	n	.41	n
12	pH			x	.05	.27	n	.20	n
13	Conductivity				x	.69	n	.63	n
14	Fe/1y/ush					x	.82	.91	.84
15	Fe/1y/sh						x	n	.98
16	Ws/1y/ush							x	.79
17	Ws/1yr/sh								x

Tab. 9 Results of stepwise regression analysis for
unsheltered exposure

Material	Factors	Regression coefficients				
		A	B1	B2	B3	R
unalloyed steel 1 year	SO ₂	179	3.4			0.75
	SO ₂ , NO ₂	180	3.5	-.07		0.76
	SO ₂ , TOW	59.3	3.8		.029	0.84
	SO ₂ , NO ₂ , TOW	79	3.8	-.08	.026	0.83
weathering steel 1 year	SO ₂	163	1.7			0.61
	SO ₂ , NO ₂	153	1.3	.58		0.65
	SO ₂ , TOW	73.4	1.9		.022	0.73
	SO ₂ , NO ₂ , TOW	69	1.6	.58	.022	0.79
weathering steel 2 years	SO ₂	231	2.1			0.56
	SO ₂ , NO ₂	222	1.8	.48		0.57
	SO ₂ , TOW	103	2.5		.031	0.74
	SO ₂ , NO ₂ , TOW	98	2.3	.36	.031	0.75
zinc 2 years	SO ₂	15.6	.20			0.44
	SO ₂ , TOW	5.3	.23		.002	0.54
	SO ₂ , NO ₂ , TOW	5.8	.26	-.04	.003	0.56

Tab.10 Results of stepwise regression analysis for sheltered exposure

Material	Factors	Regresssion coefficients				
		A	B1	B2	B3	R
unalloyed steel 1 year	SO ₂	51.2	1.6			0.56
	SO ₂ , TOW	11.2	1.8		.009	0.62
	SO ₂ , NO ₂ , TOW	25.3	2.5	-1.1	.011	0.70
weathering steel 1 year	SO ₂	68.7	1.3			0.57
	SO ₂ , TOW	28.7	1.5		.010	0.63
	SO ₂ , NO ₂ , TOW	37.3	1.8	-.63	.011	0.68
weathering steel 2 years	SO ₂	113	3.0			0.67
	SO ₂ , TOW	63.3	3.2		.011	0.70
	SO ₂ , NO ₂ , TOW	85	4.1	-1.4	.013	0.76
zinc 2 years	SO ₂	8.9	.11			0.34
	SO ₂ , TOW	4.1	.12		.001	0.40
	SO ₂ , NO ₂ , TOW	4.4	.13	-.02	.001	0.40

Tab. 11 Comparison of SO₂ deposition rate in mg/m²day parallelly measured in the open air (A) and under shelter (B)

Letnany		Pollution (mg SO ₂ /m ² .day)		Kasp. Hory	
A	B	A	B	A	B
31.5	29.4	41.1	31.0	5.5	2.2
58.6	29.4	55.0	31.0	13.1	2.2
82.3	60.8	60.2	34.7	21.8	7.0
92.7	60.8	82.7	39.3	21.5	7.0
84.2	36.1	77.9	61.8	18.2	7.2
97.5	36.1	113	61.8	25.0	7.2
90.5	36.8	81.4	45.4	17.9	11.6
68.7	36.8	73.3	45.4	19.1	11.6
39.9	23.6	62.9	31.5	8.8	4.4
46.7	23.6	53.3	31.5	8.2	4.4
22.1	15.2	42.8	21.1	3.0	2.4
29.8	15.2	49.9	21.1	3.7	2.4
35.1	33.2	31.1	37.7	5.3	2.8
57.9	33.2	47.0	37.7	11.9	2.8
99.8	38.6	81.2	48.6	23.1	7.9
102	38.6	129	48.6	23.0	7.9
94.8	35.6	100	6.3	19.9	7.2
84.6	35.6	112	6.3	20.0	7.2
57.0	34.2	77.3	3.5	14.9	7.5
64.8	34.2	95.2	3.5	7.6	7.5
37.4	18.8	79.2	38.7	12.0	7.0
32.2	18.8	50.3	38.7	6.2	7.0
28.9	19.0	45.2	26.4	4.6	2.1
32.6	19.0	60.7	26.4	6.2	2.1
44.1	22.6	47.7	29.8	11.2	4.1
32.7	18.3	51.9	29.8	9.5	4.1
64.0	27.9	145	10.4	19.9	6.2
73.2	27.9	64.1	10.4	12.0	6.2
56.9	18.6	54.4	46.2	23.9	8.1
42.3	18.6	33.8	46.2	23.8	8.1
39.1	24.4	72.7	33.1	25.1	10.4
54.7	24.4	53.1	33.1	20.1	10.4
40.4	16.5	47.8	25.9	15.4	12.7
35.1	16.5	59.4	25.9	17.4	12.7
40.3	15.3	53.3	15.1	23.6	7.6
34.5	15.3	69.0	15.1	13.6	7.6
39.5	11.7	29.8	27.0	5.2	2.3
30.2	11.7	53.4	27.0	6.9	2.3
61.3	22.3	56.2		16.2	8.1
58.9	22.3			23.2	8.1

Tab. 12 Grouping of test sites according to SO₂ and NO₂ pollution and 2 years' corrosion data (unsheltered exposure)

No.	Test site	Corrosion (g/m ²)			Environmental factors		
		Weather. steel	Zinc	Aluminium	TOW (h/year)	SO ₂ (ug/m ³)	NO ₂
<u>G1 Low polluted</u>							
35	Lahemaa	212	9.5	0.43	3849	0.4	2.4
23	Birkenes	220	20.5	0.48	4807	1.2	4.0
26	Aspvreten	180	9.0	0.46	3971	2.5	4.8
37	Dorset	173	14.2	0.84	2964	3.8	1.8
29	Clatteringshaws	267	23.7	1.06		4.3	3.4
5	Ahtari	159	8.8	0.48	2985	4.5	5.0
33	Toledo	93.1	4.5	0.23	2062	6.2	11.8
12	Garmisch-Part.	162	12.6	0.29	4983	11.5	12.9
7	Waldhof	315	26.9	0.56	4214	13.3	12.5
4	Espoo	275	15.5	0.71	3020	15.4	19.6
2	Kasp. Hory	236	14.0	0.48	3351	17.8	15.6
<u>G2 Moderate polluted</u>							
28	Wells Cath.	280	16.0	0.77	5670	7.0	23.2
18	Eibergen	263	12.6	0.42	5370	9.1	25.0
36	Lisbon	256	14.2	0.86	1575	9.9	29.1
39	Res. Tri. Park	218	19.0	0.54	3178	10.4	27.8
19	Vredepeel	317	18.5	0.63	5318	11.4	31.4
30	Stoke Orchard	333	23.1	1.75	4075	12.0	57.9
20	Wijnandsrade	278	20.0	0.49	5167	12.8	30.2
21	Oslo	242	16.2	0.99	2627	13.5	51.8
24	Stockholm S	299	23.2	1.16	3805	14.5	28.9
31	Madrid	199	9.6	0.40	1868	18.2	28.2
27	Lincoln Cath.	424	17.0	2.73	5907	18.8	61.4
6	Helsinki	356	20.6	1.69	3633	19.1	29.3
8	Aschaffenburg	262	40.1	0.56	3756	19.1	39.0
25	Stockholm C	329	14.4	1.01	3805	20.4	45.7
16	Venice	273	18.3	0.54	4205	20.5	39.5
34	Moscow	250	14.0	0.39	2383	22.4	72.3
9	Langenfeld	334	42.6	0.96	4487	25.4	46.4
11	Essen	342	23.4	1.36	4624	29.1	45.6
17	Vlaardingen	347	25.9	1.11	5732	33.5	54.3
13	Rome	204	11.8	0.40	1667	34.1	73.2
<u>G3 Medium to heavy polluted</u>							
32	Bilbao	295	25.2	0.87	4233	42.5	38.9
22	Borregaard	430	48.3	3.80	3255	44.8	18.3
10	Bottrop	380			4336	49.8	48.8
38	Steubenville	280	20.1	1.10	1946	58.8	43.3
1	Letnany	361	26.1	0.91	3006	75.9	37.5
15	Milan	316	23.1	0.96	3454	76.2	109
3	Kopisty	427	29.6	1.25	2377	89.0	40.7

Tab. 13 Grouping test sites according to SO₂ and NO₂ pollution and 2 years' corrosion data (sheltered exposure)

No.	Test site	Corrosion (g/m ²)			Environmental factors		
		Weather. steel	Zinc	Aluminium	TOW (h/year)	SO ₂ (ug/m ³)	NO ₂
<u>G1 Low polluted</u>							
35	Lahemaa	156	6.2	1.09	3849	0.4	2.4
23	Birkenes	103	12.1	0.34	4807	1.2	4.0
26	Aspvreten	105	5.7	0.28	3971	2.5	4.8
37	Dorset	118	6.0	0.74	2964	3.8	1.8
29	Clatteringshaws	223	7.0	1.32		4.3	3.4
5	Ahtari	97.0	6.7	0.35	2985	4.5	5.0
33	Toledo	50.3	2.3	0.21	2062	6.2	11.8
12	Garmisch-Part.	115	5.6	0.28	4983	11.5	12.9
7	Waldhof	190	21.6	0.60	4214	13.3	12.5
4	Espoo	158	10.1	0.68	3020	15.4	19.6
2	Kasp. Hory	180	8.8	0.90	3351	17.8	15.6
<u>G2 Moderate polluted</u>							
28	Wells Cath.	162	7.8	1.59	5670	7.0	23.2
18	Eibergen	127	5.8	0.40	5370	9.1	25.0
36	Lisbon	141	6.2	1.34	1575	9.9	29.1
39	Res. Tri. Park	176	7.4	0.92	3178	10.4	27.8
19	Vredepeel	173	7.4	1.05	5318	11.4	31.4
30	Stoke Orchard	235	9.5	2.69	4075	12.0	57.9
20	Wijnandsrade	134	6.5	0.49	5167	12.8	30.2
21	Oslo	121	7.1	0.81	2627	13.5	51.8
24	Stockholm S	150	19.8	0.98	3805	14.5	28.9
31	Madrid	109	8.0	0.45	1868	18.2	28.2
27	Lincoln Cath.	251	10.1	2.86	5907	18.8	61.4
6	Helsinki	205	15.1	1.57	3633	19.1	29.3
8	Aschaffenburg	133	22.2	0.56	3756	19.1	39.0
25	Stockholm C	123	7.5	0.90	3805	20.4	45.7
16	Venice	161	9.3	1.21	4205	20.5	39.5
34	Moscow	120	10.8	0.59	2383	22.4	72.3
9	Langenfeld	182	31.8	1.07	4487	25.4	46.4
11	Essen	204	9.8	2.76	4624	29.1	45.6
17	Vlaardingen	230	18.3	1.92	5732	33.5	54.3
13	Rome	106	7.7	0.75	1667	34.1	73.2
<u>G3 Medium to heavy polluted</u>							
32	Bilbao	118	7.8	1.22	4233	42.5	38.9
22	Borregaard	510	25.4	5.30	3255	44.8	18.3
10	Bottrop	201	20.2	0.96	4336	49.8	48.8
38	Steubenville				1946	58.8	43.3
1	Letnany	365	13.3	1.71	3006	75.9	37.5
15	Milan	206	10.6	1.35	3454	76.2	109
3	Kopisty	476	14.3	1.98	2377	89.0	40.7

Fig. 1 Corrosion of unalloyed steel (mass losses) after 1 year of exposure

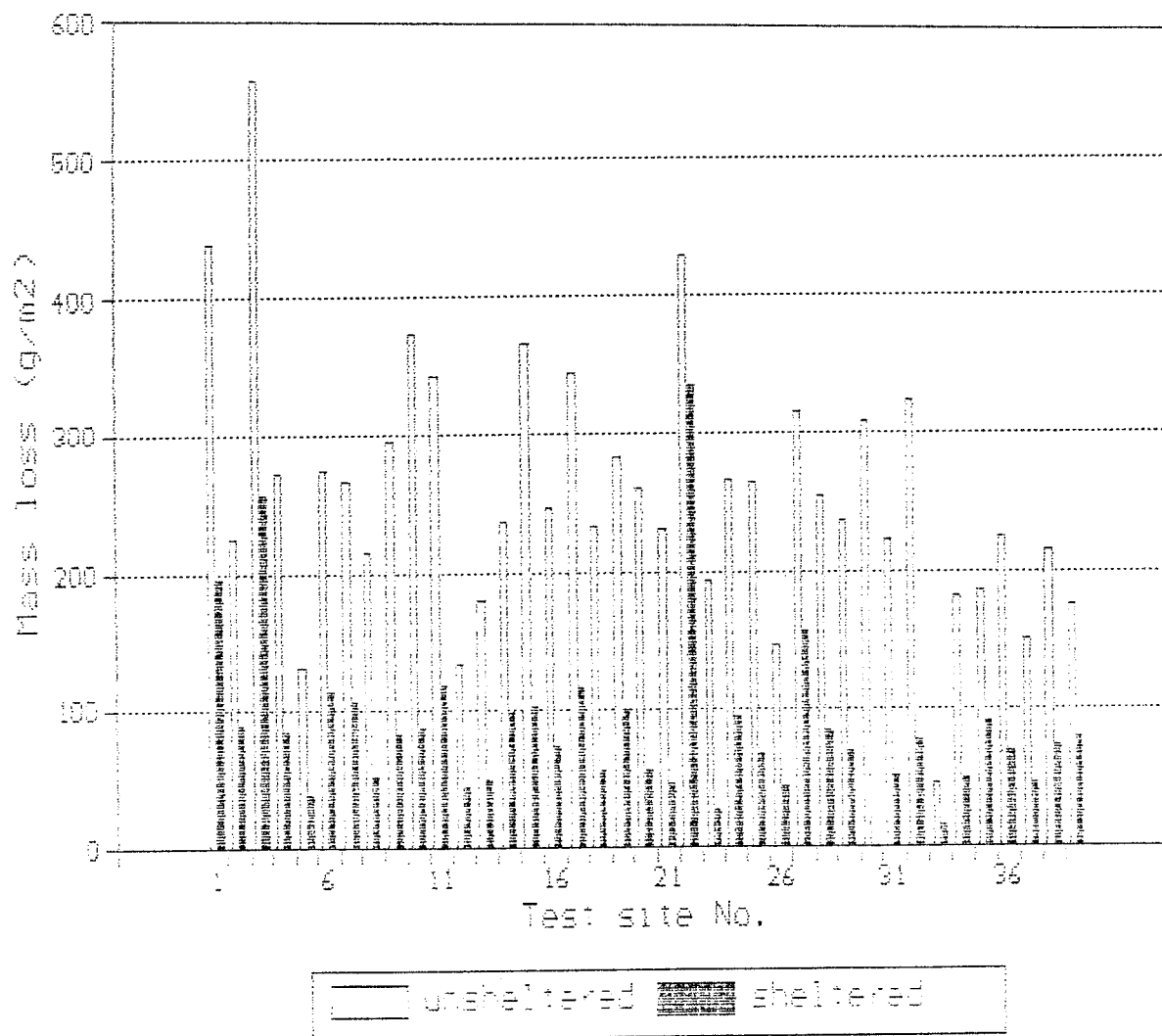


Fig. 2 Corrosion of weathering steel Atmofix (mass losses) after 1 year of exposure

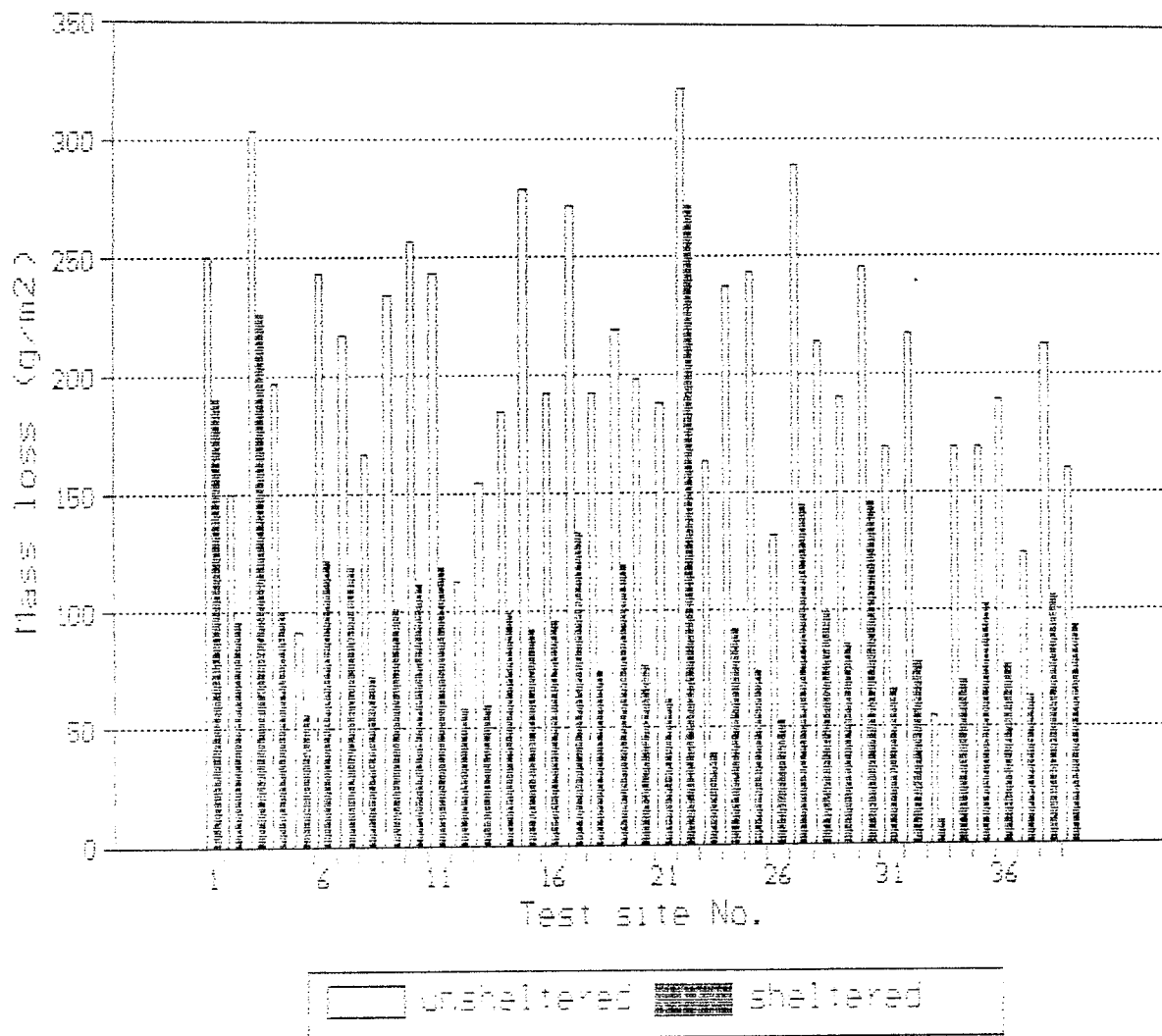


Fig. 3 Corrosion of weathering steel Atmofix (mass losses) after 2 years of exposure

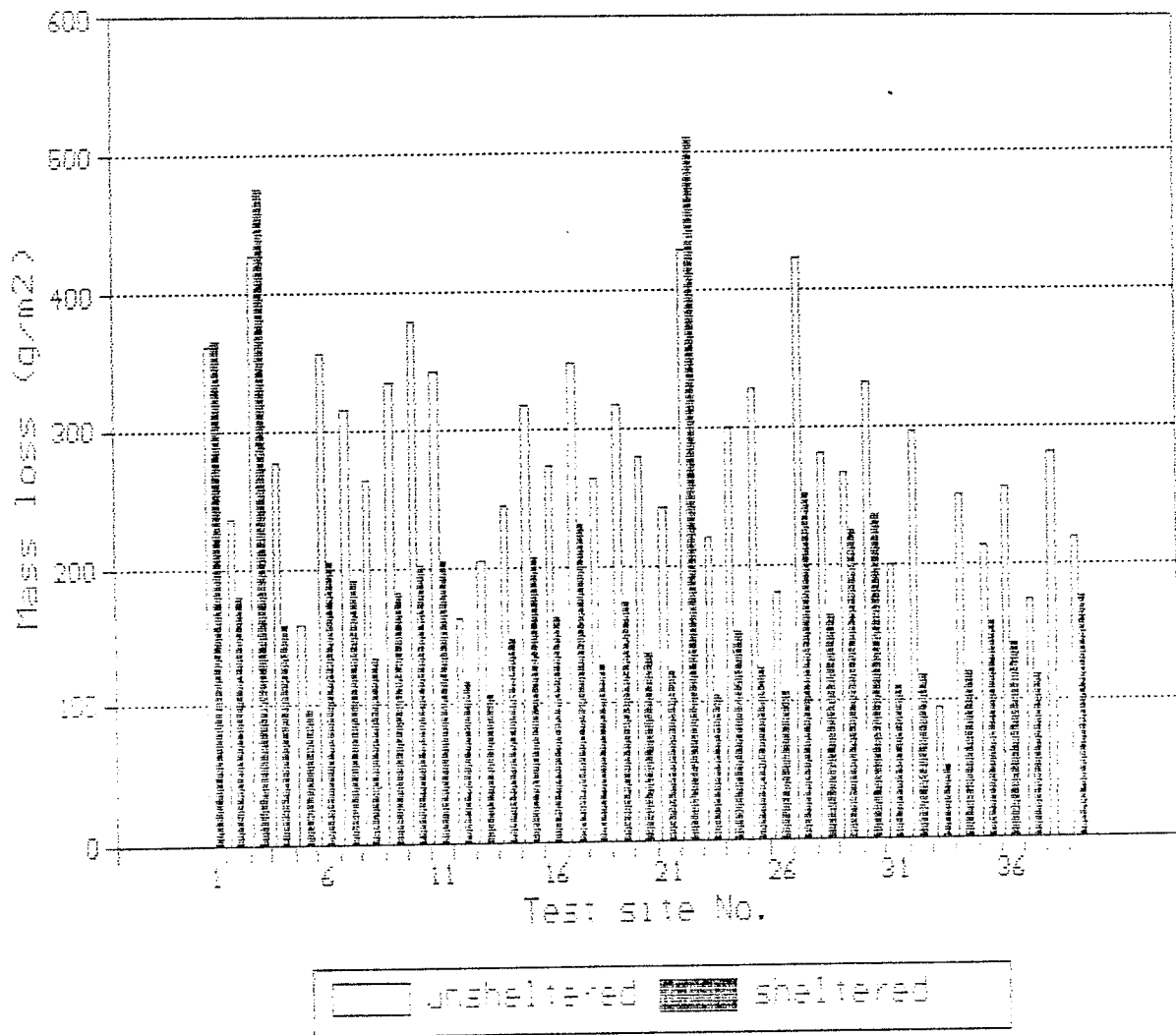


Fig. 4 Corrosion of zinc (mass losses) after 2 years of exposure

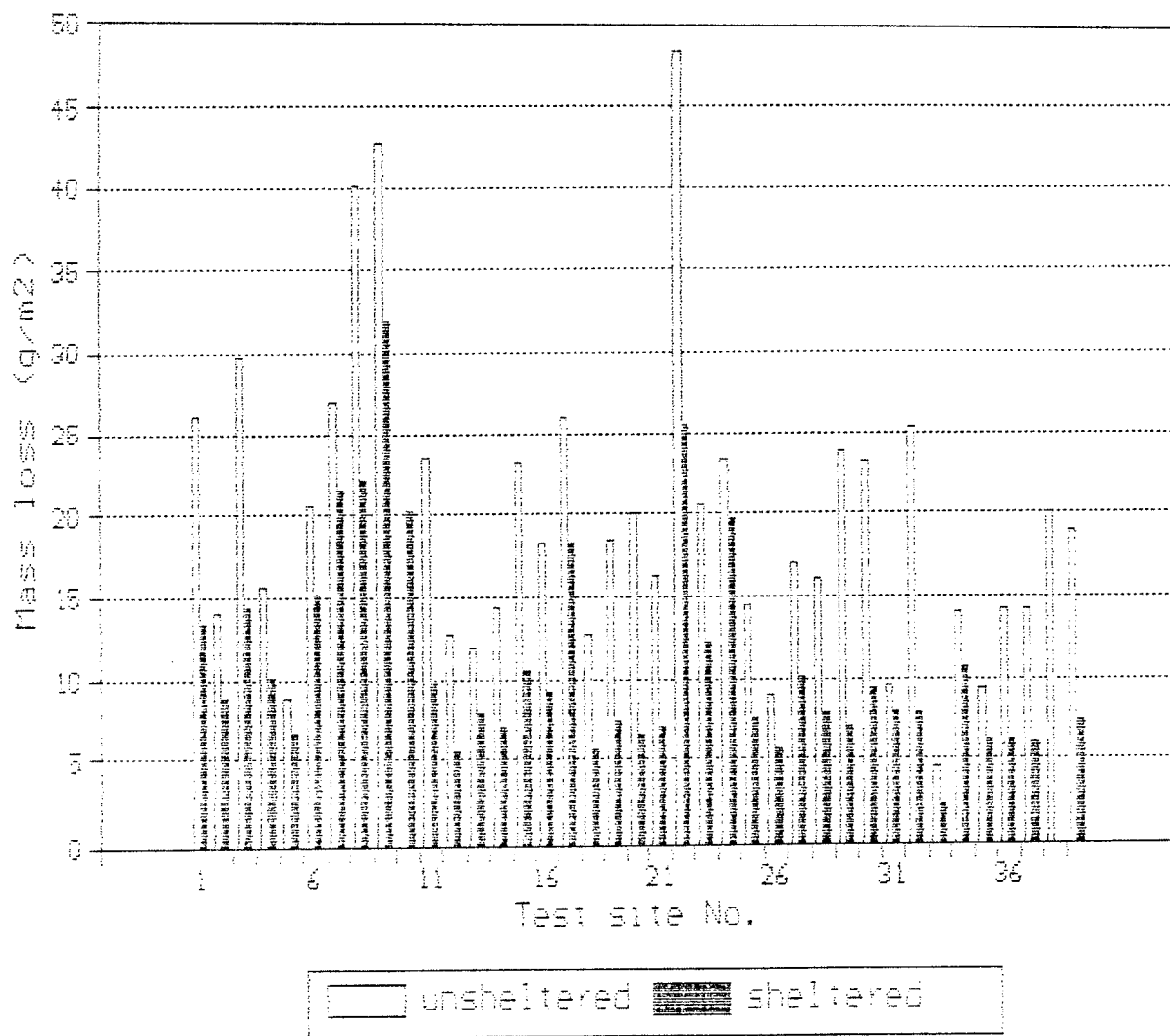


Fig. 5 Corrosion of aluminium (mass losses) after 2 years of exposure

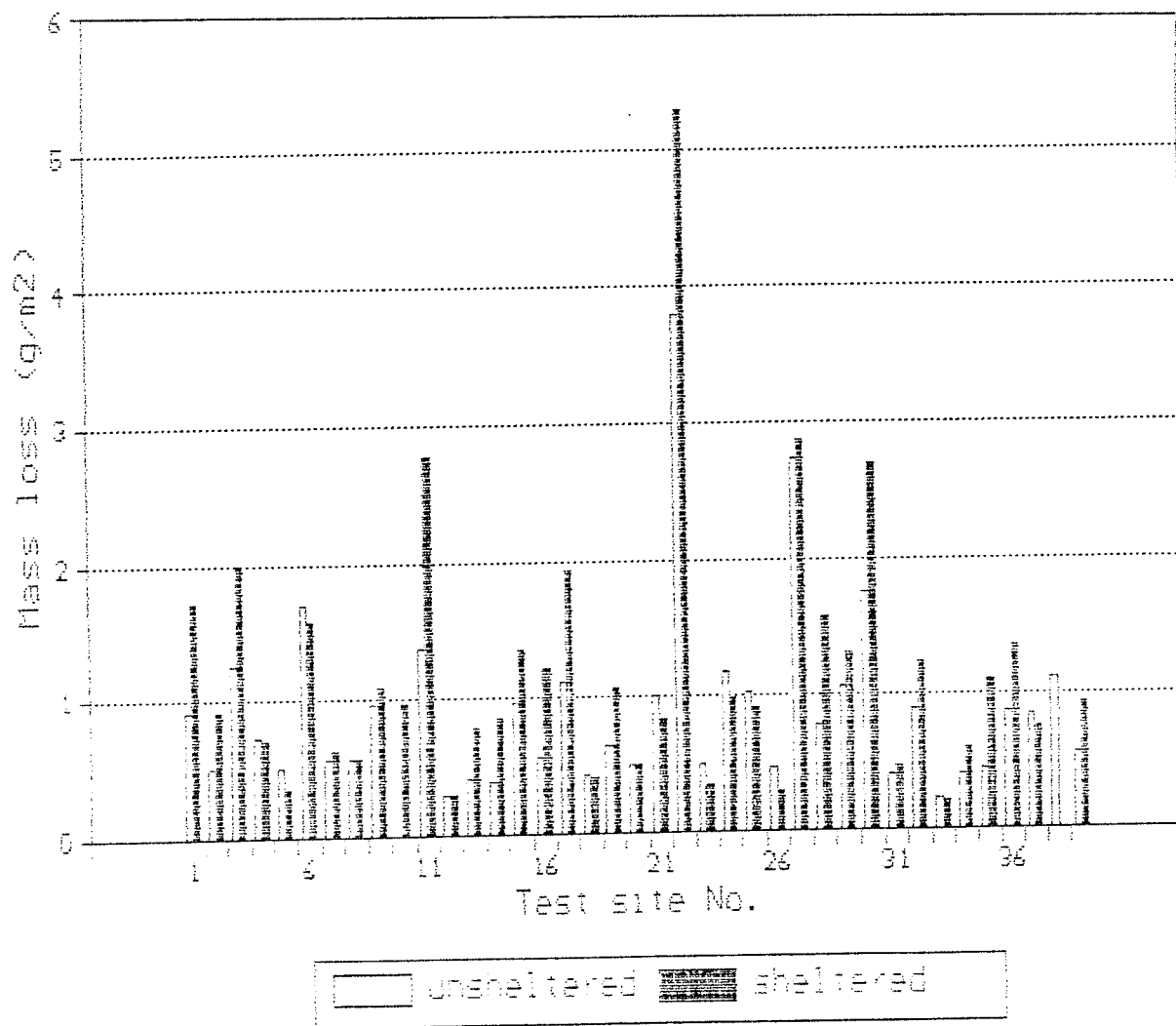
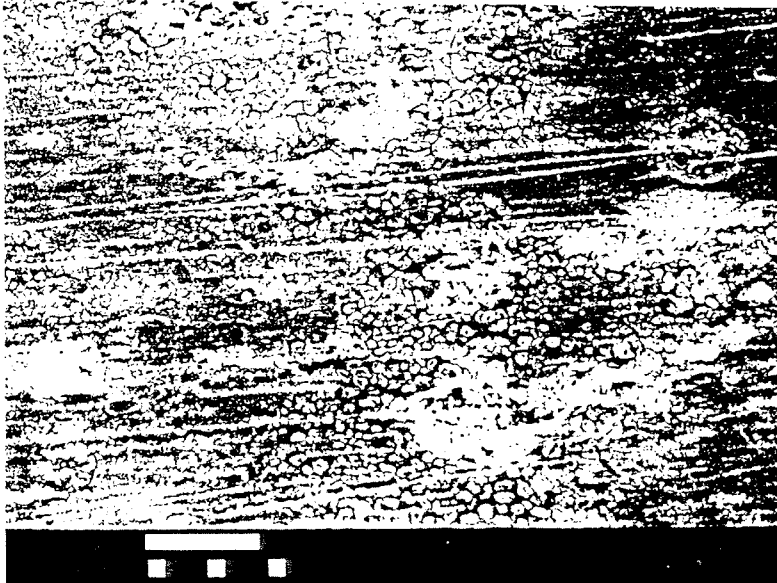


Fig. 6 Local corrosion of aluminium after 2 years of exposure (SEM pictures of corroded surface after pickling)
test site No. 22 Borregaard
a - unsheltered, skyward side (150X)
b - unsheltered, groundward side (150X)
c - sheltered (150X)



a



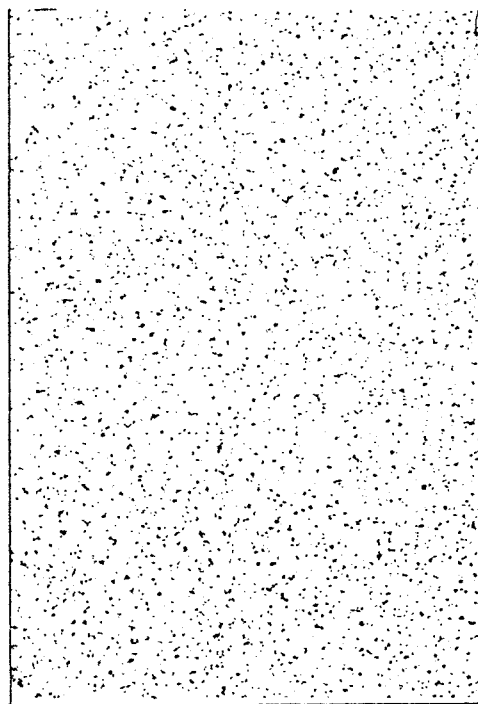
b



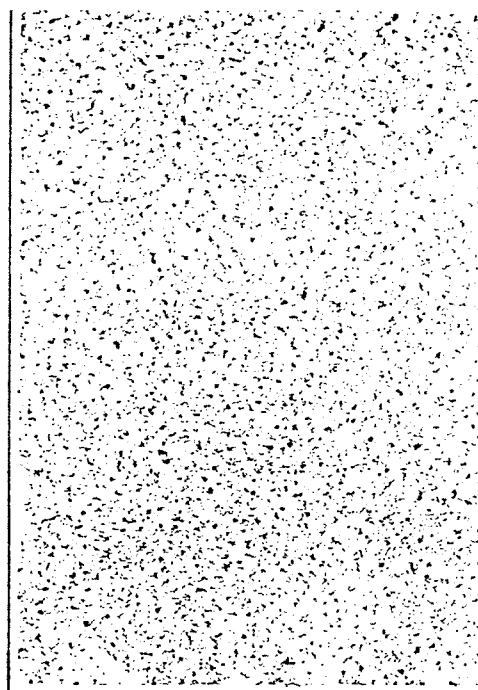
c

Fig. 7 Amount of non-adherent rust ("rust prints") on weathering steel Atmofix after 2 years of exposure
A - No. 5 Ahtari(rural) (a-unsheltered, b-sheltered)
B - No. 28 Wells Cathedral (urban)
C - No. 22 Borregaard (industrial)
D - No. 33 Toledo (rural dry atmosphere)

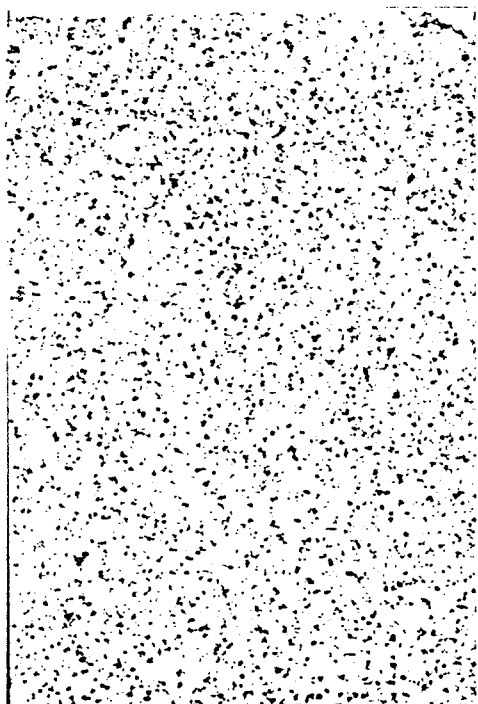
Aa



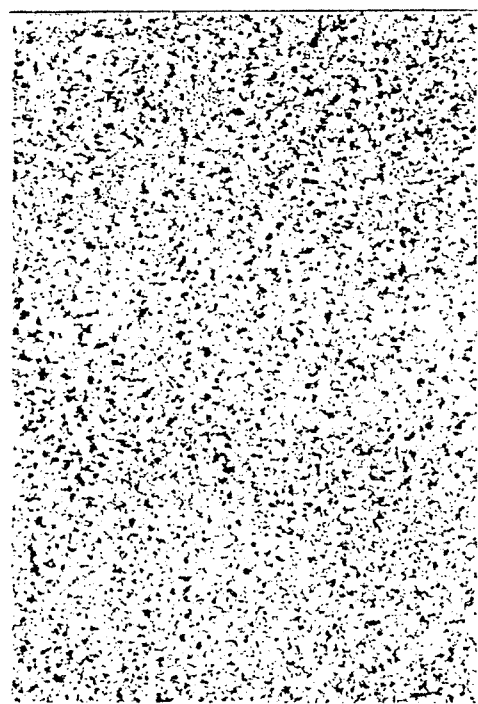
Ab



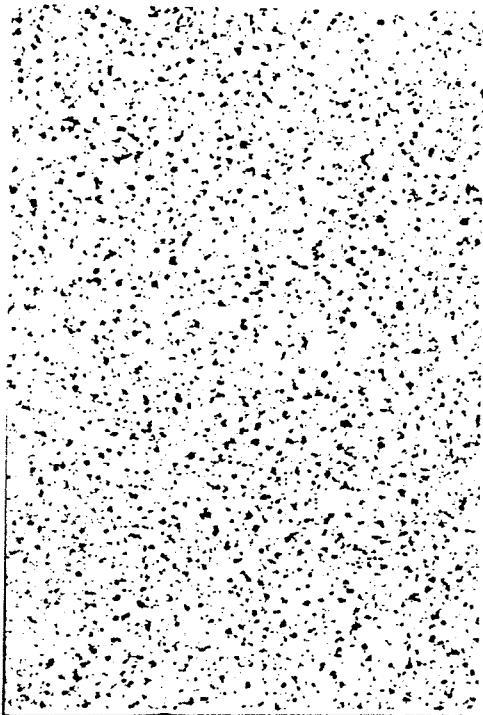
Ba



Bb



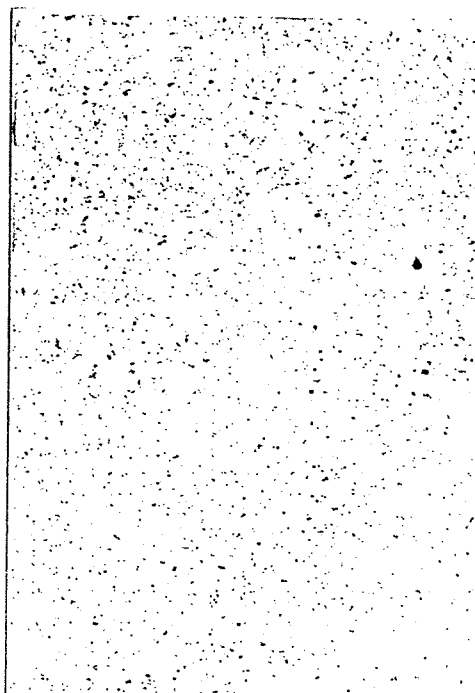
Ca



Cb



Da



Db



Fig. 8 Comparison of corrosion of unalloyed and weathering steel Atmofix after 1 year of exposure (unsheltered)

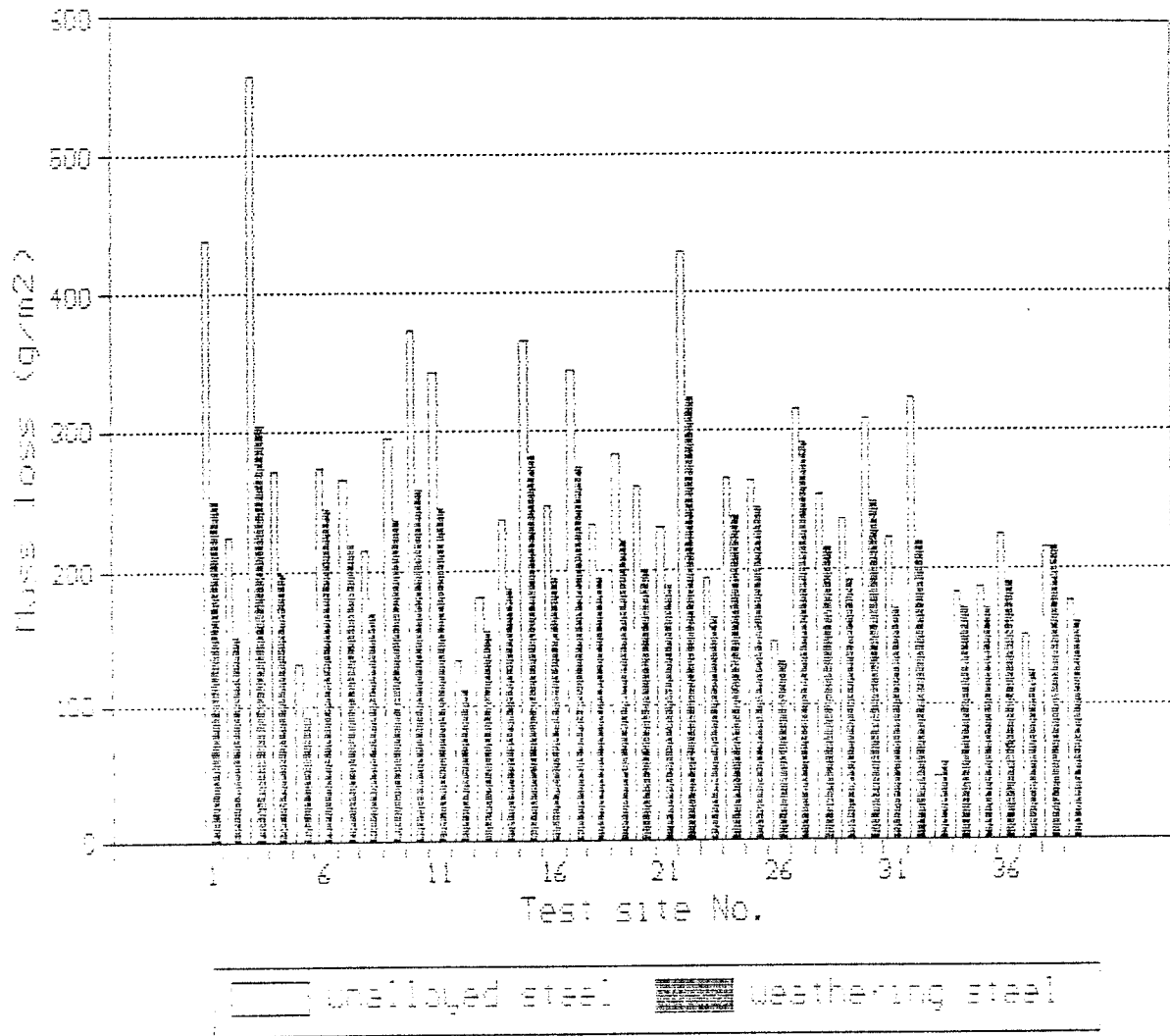


Fig. 9 Mass losses of weathering steel Atmofix after 2 years of unsheltered exposure vs. SO₂ pollution

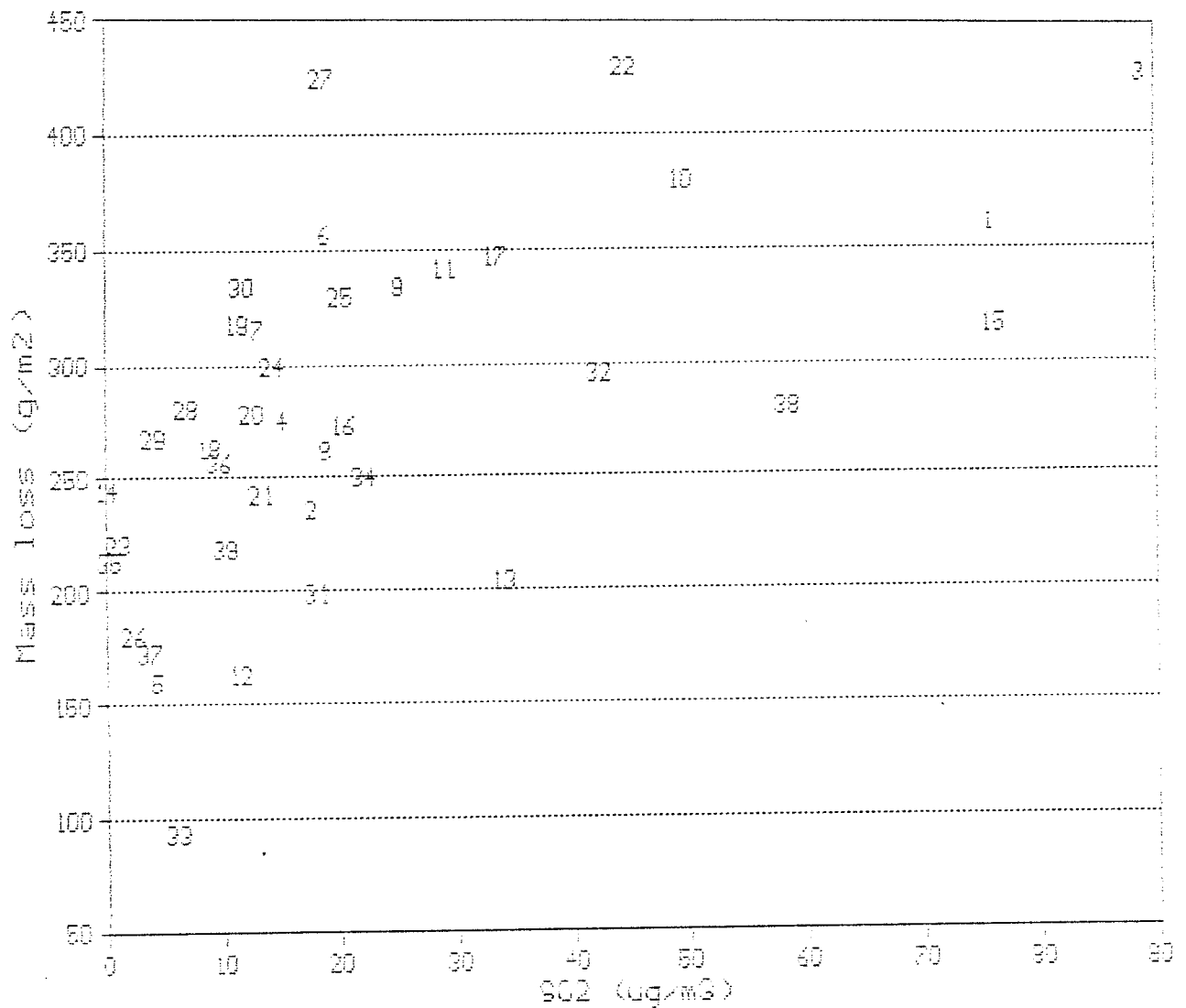


Fig. 10 Mass losses of zinc after 2 years of unsheltered exposure vs. SO_2 pollution

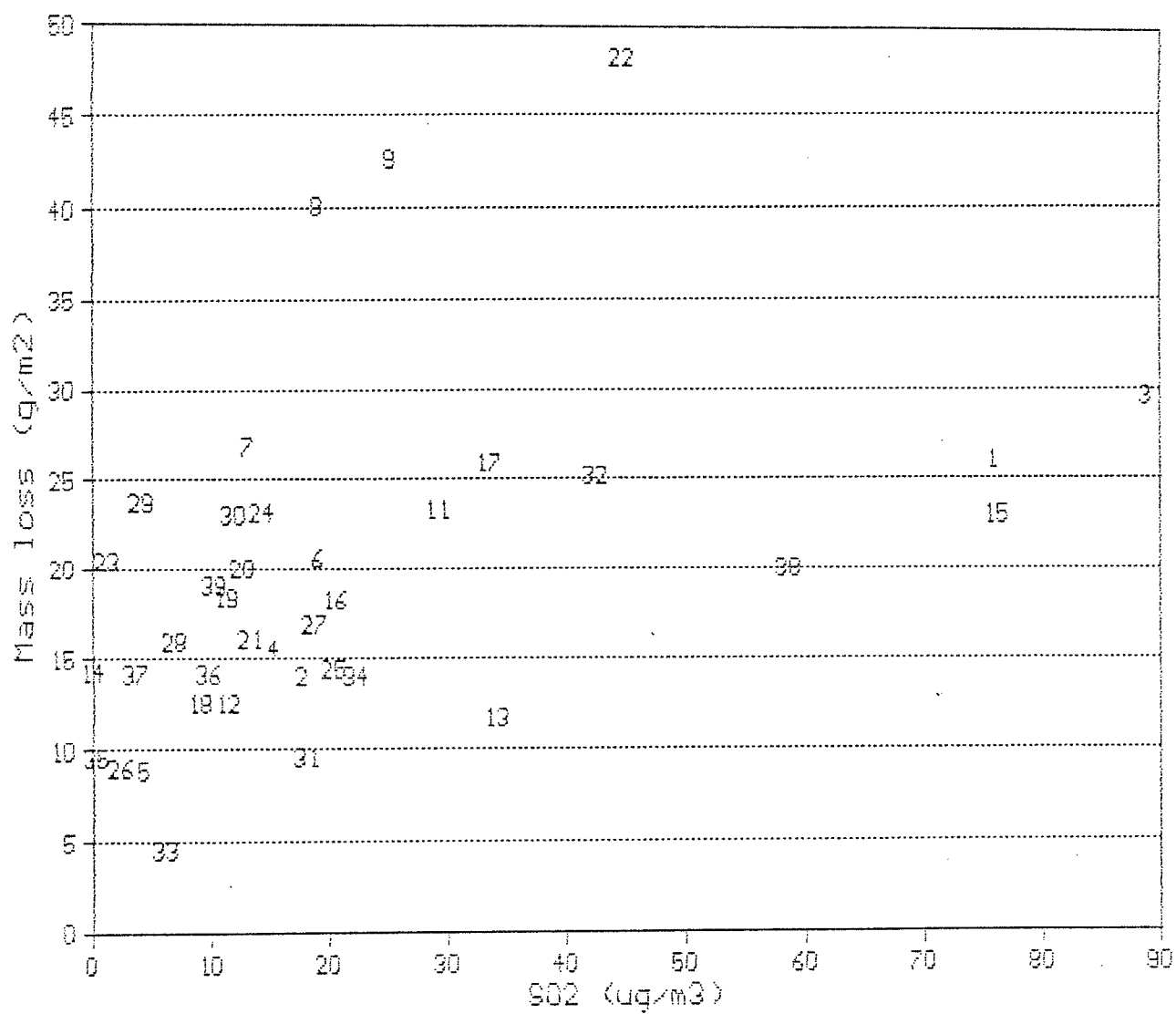


Fig. 11 Mass losses of aluminium after 2 years of unsheltered exposure vs. SO_2 pollution

